CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 10: Arithmetic Progression

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Sum of n terms

1. An A.P. has first term 5 and last term 50, and its terms add up to 440. Find (a) the number of terms (b) the common difference. ICSE 2025, 3 marks

SOLUTION

(a) $S = \frac{n}{2}(a + l)$: $440 = \frac{n}{2}(5 + 50)$, so $n = \frac{880}{55} = 16$.

(b) $l = a + (n - 1)d$: $50 = 5 + 15d$, so $d = 3$.

Answer: (a) 16 terms (b) common difference 3

2. The nth term of an A.P. is $T_n = 6(7 - n)$. Find (a) its first term and common difference (b) the sum of its first 25 terms. ICSE 2024, 3 marks

SOLUTION

(a) $T_1 = 6 \times 6 = 36$ and $T_2 = 6 \times 5 = 30$, so $a = 36$ and $d = -6$.

(b) $S_{25} = \frac{25}{2}[2(36) + 24(-6)] = \frac{25}{2} \times (-72)$

Answer: (a) $a = 36$, $d = -6$ **(b)** $S_{25} = -900$

3. The 5th term of an A.P. is 4 and its 9th term is -12. Find (a) the first term (b) the common difference (c) the sum of the first 16 terms. ICSE 2023, 3 marks

SOLUTION

$a + 4d = 4$ and $a + 8d = -12$. Subtracting: $4d = -16$, so $d = -4$ and $a = 20$.

$S_{16} = \frac{16}{2}[2(20) + 15(-4)] = 8 \times (-20)$

Answer: (a) 20 (b) -4 **(c)** -160

4. Which term of the A.P. $15, 30, 45, 60, \ldots$ is 300? Find the sum of all the terms up to it. ICSE 2023, 3 marks

SOLUTION

$a = 15$, $d = 15$. $15 + (n - 1) \times 15 = 300$, so $15n = 300$ and $n = 20$.

$S_{20} = \frac{20}{2}(15 + 300) = 10 \times 315$

Answer: the 20th term; sum $= 3150$

5. The 6th term of an A.P. is four times its first term, and the sum of its first six terms is 75. Find the first term and the common difference. ICSE 2020, 3 marks

SOLUTION

$a + 5d = 4a$, so $5d = 3a$.

$S_6 = \frac{6}{2}(2a + 5d) = 3(2a + 3a) = 15a = 75$, so $a = 5$.

$5d = 15$, so $d = 3$.

Answer: first term 5, common difference 3

6. The 4th and 6th terms of an A.P. are 8 and 14. Find (a) the first term (b) the common difference (c) the sum of the first 20 terms. ICSE 2019, 4 marks

SOLUTION

$a + 3d = 8$ and $a + 5d = 14$. Subtracting: $2d = 6$, so $d = 3$ and $a = -1$.

$S_{20} = \frac{20}{2}[2(-1) + 19(3)] = 10 \times 55$

Answer: (a) -1 (b) 3 (c) 550

7. The 4th term of an A.P. is 22 and the 15th term is 66. Find the first term and the common difference, and then the sum of the first 8 terms. ICSE 2018, 4 marks

SOLUTION

$a + 3d = 22$ and $a + 14d = 66$. Subtracting: $11d = 44$, so $d = 4$ and $a = 10$.

$S_8 = \frac{8}{2}[2(10) + 7(4)] = 4 \times 48$

Answer: $a = 10$, $d = 4$, $S_8 = 192$

2 The general term

8. Find the 7th term of the A.P. $\frac{1}{a}, \frac{1}{a} + 1, \frac{1}{a} + 2, \ldots$ (a) $\frac{1}{a} + 6$ (b) $\frac{1}{a} + 7$ (c) $\frac{1}{a} + 8$ (d) $\frac{1}{a} + 7^7$

ICSE 2024, 1 mark

SOLUTION

First term $= \frac{1}{a}$ and $d = 1$. $T_7 = \frac{1}{a} + 6 \times 1$

Answer: (a)

9. The nth term of an A.P. is $2n + 5$. Find its 10th term. (a) 7 (b) 15 (c) 25 (d) 45 ICSE 2023, 1 mark

SOLUTION

$T_{10} = 2(10) + 5 = 25$

Answer: (c)

10. $(k-3)$, $(2k+1)$ and $(4k+3)$ are three consecutive terms of an A.P. Find k. ICSE 2018, 3 marks

SOLUTION

The middle term is the average of the other two: $2(2k+1) = (k-3) + (4k+3)$

$4k + 2 = 5k$

Answer: $k = 2$

3 Choosing terms conveniently

11. The first three terms of an A.P. add up to 42, and the product of the first and third terms is 52. Find the first term and the common difference. ICSE 2019, 3 marks

SOLUTION

Let the terms be $a - d$, a, $a + d$. Their sum is $3a = 42$, so $a = 14$.

$(14 - d)(14 + d) = 52$, so $d^2 = 144$ and $d = \pm 12$.

Answer: first term 2 with common difference 12, or first term 26 with common difference -12

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